

STAT 9120: Theory of Linear Models

Homework: Quadratic Forms, Cochran's Theorem, and Hypothesis Testing

Fall 2026

Instructions. Show all work and justify your answers. You may use software to check numerical calculations, but your submitted solution should demonstrate the relevant matrix and distributional arguments.

Unless otherwise stated, assume the normal linear model

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim N_n(0, \sigma^2 I).$$

Problem 1. Distribution of a General Quadratic Form

12 points

Let

$$Z \sim N_3(0, I_3)$$

and consider

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Define

$$Q = Z^T A Z.$$

(a) Find the eigenvalues of A .

[3 points]

(b) Using the spectral decomposition of A , show that

$$Q \stackrel{d}{=} 3U_1^2 + U_2^2,$$

where U_1, U_2 are independent standard normal random variables.

[3 points]

(c) Is Q chi-squared distributed? Explain why or why not.

[3 points]

(d) Find $E(Q)$ and $\text{Var}(Q)$.

[3 points]

Problem 2. Projection Quadratic Forms and the Noncentral Chi-Squared Distribution

12 points

Consider

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Suppose

$$Y \sim N_3(\mu, \sigma^2 I).$$

(a) Verify that P is symmetric and idempotent, and determine $\text{rank}(P)$.

[3 points]

- (b) State the distribution of

$$\frac{Y^T P Y}{\sigma^2}.$$

Give its degrees of freedom and noncentrality parameter.

[3 points]

- (c) Suppose

$$\mu = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \quad \sigma^2 = 2.$$

Find the noncentrality parameter and state the distribution explicitly.

[3 points]

- (d) Under what condition on
- μ
- does

$$\frac{Y^T P Y}{\sigma^2}$$

have a **central** chi-squared distribution? Give a geometric interpretation.

[3 points]

Problem 3. Cochran's Theorem and the Regression Decomposition

12 points

Suppose $X \in \mathbb{R}^{n \times p}$ has full column rank and contains an intercept. Define

$$H = X(X^T X)^{-1} X^T$$

and

$$J = \frac{1}{n} \mathbf{1} \mathbf{1}^T.$$

Consider the two matrices

$$P_R = H - J, \quad P_E = I - H.$$

- (a) Show that P_R and P_E are symmetric and idempotent. You may use the fact that, because the model contains an intercept,

$$HJ = JH = J.$$

[3 points]

- (b) Show that

$$P_R P_E = 0.$$

Explain what this means geometrically.

[3 points]

- (c) Show that

$$\text{rank}(P_R) = p - 1 \quad \text{and} \quad \text{rank}(P_E) = n - p.$$

[3 points]

- (d) Under the global null hypothesis that all slope coefficients are zero, use Cochran's theorem to establish that

$$\frac{Y^T (H - J) Y}{\sigma^2} \sim \chi_{p-1}^2, \quad \frac{Y^T (I - H) Y}{\sigma^2} \sim \chi_{n-p}^2,$$

and that the two quantities are independent.

[3 points]

Problem 4. The Residual Sum of Squares**10 points**

Suppose the normal linear model has

$$n = 30, \quad p = 5.$$

Let

$$SSE = Y^T(I - H)Y$$

and

$$MSE = \frac{SSE}{n - p}.$$

- (a) State the exact distribution of

$$\frac{SSE}{\sigma^2}.$$

Explain why the degrees of freedom are 25.

[3 points]

- (b) Use the mean and variance of a chi-squared random variable to determine

$$E(MSE) \quad \text{and} \quad \text{Var}(MSE).$$

[4 points]

- (c) Explain geometrically why estimating p regression parameters leaves $n - p$ residual degrees of freedom. Your explanation should refer to the dimension of

$$\mathcal{C}(X)^\perp.$$

[3 points]**Problem 5. Nested Models and the Partial F -Test****16 points**

Consider two nested models,

$$\mathcal{C}(X_0) \subset \mathcal{C}(X_1),$$

with

$$n = 50, \quad p_0 = 3, \quad p_1 = 6.$$

Suppose the residual sums of squares are

$$SSE_0 = 120$$

for the reduced model and

$$SSE_1 = 90$$

for the full model. Let H_0 and H_1 denote the corresponding projection matrices.

- (a) Explain why $H_1 - H_0$ is an orthogonal projection matrix and show that

$$\text{rank}(H_1 - H_0) = p_1 - p_0 = 3.$$

[4 points]

- (b) Under the null hypothesis that the reduced model is correct, state the distribution of

$$\frac{SSE_0 - SSE_1}{\sigma^2}.$$

[3 points]

- (c) Explain why $SSE_0 - SSE_1$ and SSE_1 are independent under the null hypothesis. Your argument should refer to the projection matrices

$$H_1 - H_0 \quad \text{and} \quad I - H_1.$$

[3 points]

- (d) Compute the partial F -statistic

$$F = \frac{(SSE_0 - SSE_1)/(p_1 - p_0)}{SSE_1/(n - p_1)}.$$

At significance level $\alpha = 0.05$, reject H_0 if

$$F > 2.82.$$

State your conclusion in statistical terms.

[6 points]

Problem 6. A General Linear Hypothesis

16 points

Consider a normal linear model with

$$n = 25, \quad p = 3,$$

where the first column of X is an intercept. Suppose

$$\hat{\beta} = \begin{pmatrix} 2 \\ 1 \\ -0.5 \end{pmatrix},$$

$$(X^T X)^{-1} = \begin{pmatrix} 0.20 & 0 & 0 \\ 0 & 0.25 & 0.05 \\ 0 & 0.05 & 0.20 \end{pmatrix},$$

and

$$MSE = 1.$$

We wish to test

$$H_0 : \beta_2 = \beta_3 = 0.$$

- (a) Write the hypothesis in the form

$$H_0 : C\beta = d.$$

Specify C , d , and

$$q = \text{rank}(C).$$

[3 points]

- (b) Compute

$$C\hat{\beta} - d$$

and

$$C(X^T X)^{-1}C^T.$$

Then calculate the quadratic form

$$Q = (C\hat{\beta} - d)^T [C(X^T X)^{-1}C^T]^{-1} (C\hat{\beta} - d).$$

[5 points]

(c) Compute

$$F = \frac{Q/q}{MSE}.$$

The 0.05-level critical value is

$$F_{2,22;0.95} = 3.44.$$

State your conclusion.

[4 points]

(d) Interpret this test geometrically as a comparison between a reduced model space and a full model space. What additional directions are being tested?

[4 points]

Problem 7. The Relationship Between t - and F -Tests

10 points

Suppose we wish to test the single linear hypothesis

$$H_0 : c^T \beta = d.$$

The usual t -statistic is

$$t = \frac{c^T \hat{\beta} - d}{\sqrt{MSE c^T (X^T X)^{-1} c}}.$$

(a) Starting from the general linear F -statistic with $q = 1$, show that

$$\boxed{F = t^2}.$$

[4 points]

(b) Suppose

$$c^T \hat{\beta} - d = 1.2, \quad c^T (X^T X)^{-1} c = 0.16, \quad MSE = 1.$$

Compute both t and F .

[3 points]

(c) Suppose the relevant critical values are

$$t_{22;0.975} = 2.074 \quad \text{and} \quad F_{1,22;0.95} = 4.30.$$

Verify that the two tests give the same decision. Explain why they must always agree for a two-sided test of a single linear hypothesis.

[3 points]

Problem 8. Hypothesis Testing in a Rank-Deficient Model

12 points

Consider the effects model

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad i = 1, 2,$$

with two observations per group. Using the parameter vector

$$\beta = \begin{pmatrix} \mu \\ \alpha_1 \\ \alpha_2 \end{pmatrix},$$

the design matrix is

$$X = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}.$$

- (a) Find a nonzero vector
- h
- satisfying

$$Xh = 0.$$

Hence explain why the model is rank deficient.

[3 points]

- (b) Determine which of the following hypotheses involve estimable—and therefore testable—functions:

$$H_{01} : \mu = 0, \quad H_{02} : \alpha_1 = 0, \quad H_{03} : \alpha_1 - \alpha_2 = 0, \quad H_{04} : \mu + \alpha_1 = 0.$$

Justify your answers using the condition

$$c^T h = 0.$$

[4 points]

- (c) Show that testing equality of the two group means can be expressed as

$$H_0 : \alpha_1 - \alpha_2 = 0.$$

Why is this hypothesis meaningful even though α_1 and α_2 are not individually estimable? [3 points]

- (d) Explain why imposing an identifying constraint such as

$$\alpha_1 + \alpha_2 = 0$$

may change the numerical values assigned to the individual parameters but cannot change the result of a valid test of equality of the group means. [2 points]

Total: 100 points

Key Results to Review

For

$$Z \sim N_n(0, I)$$

and a symmetric matrix A with eigenvalues $\lambda_1, \dots, \lambda_n$,

$$Z^T A Z \stackrel{d}{=} \sum_{j=1}^n \lambda_j Z_j^2.$$

If P is a symmetric idempotent matrix of rank r ,

$$Z^T P Z \sim \chi_r^2.$$

For

$$Y \sim N_n(\mu, \sigma^2 I),$$

$$\frac{Y^T P Y}{\sigma^2} \sim \chi_r^2(\lambda), \quad \lambda = \frac{\mu^T P \mu}{\sigma^2}.$$

Cochran's theorem provides the independence of quadratic forms corresponding to mutually orthogonal projection matrices.

For nested models,

$$\mathcal{C}(X_0) \subset \mathcal{C}(X_1),$$

the additional model space is represented by

$$H_1 - H_0,$$

and

$$F = \frac{(SSE_0 - SSE_1)/(p_1 - p_0)}{SSE_1/(n - p_1)} \sim F_{p_1 - p_0, n - p_1}$$

under the null.

More generally, for

$$H_0 : C\beta = d,$$

$$F = \frac{(C\hat{\beta} - d)^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta} - d)/q}{MSE} \sim F_{q, n-p}$$

under H_0 , provided the q restrictions are linearly independent and testable.

For a single restriction,

$$F = t^2.$$